

Day 03 – Unit 3 : Development and Planning

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Round I - Deep Learning

1. Measurement of Development

> HDI - Human Development Index

- UNDP, pehli HDR **1990**. Architects — Mahbub ul Haq + Amartya Sen
- Idea: development = "expanding people's choices", sirf GDP nahi
- 3 dimensions, 4 indicators: Health (Life Expectancy, **20-85**) · Education (MYS **0-15**, EYS **0-18**) · Living Std. (GNI pc PPP, **100-75,000**)
- Dimension Index = (Actual - Min) / (Max - Min) — income mein **log** lagta hai
- Education index = MYS & EYS ka arithmetic mean

$$HDI = (I_{Health} \times I_{Edu} \times I_{Income})^{1/3} = \text{GEOMETRIC MEAN}$$

EXAM POINT: Geometric mean **2010 se** (pehle arithmetic tha). Reason — **poor substitutability**: ek dimension ki kharaabi doosre se compensate nahi hoti.

Very High	High	Medium	Low
≥ 0.800	0.700 – 0.799	0.550 – 0.699	< 0.550

- India (HDR 2025): Rank **130 / 193**, HDI **0.685**, Medium (pichhli baar 134 / 0.644)

> GDI & GII

Point	GDI - Gender Development Index	GI - Gender Inequality Index
Intro	HDR 1995 (revised 2014)	HDR 2010 — GEM ko replace kiya
Formula	$GDI = \text{Female HDI} \div \text{Male HDI}$ (ratio hai, index nahi)	Geometric mean → phir dono genders ka harmonic mean
Dimensions	Wahi 3 HDI dimensions, gender-wise alag	Reproductive Health (MMR + ABR) · Empowerment (Parl. seats + Sec. edu.) · Labour Mkt (LFPR)
Reading	1 = parity < 1 = female peeche. India Group 5	0 = equality , 1 = inequality → kam value = behtar. India ~0.403

Do sabse bade traps:

1. **GII mein income / GNI hai hi nahi** — uski jagah **LFPR** hai.
2. **GII ki direction ulta hai** — HDI/GDI mein zyada = achha, GII mein kam = achha.
3. Reproductive health ka **koi male counterpart nahi** — yahi GII ki badi criticism.

- **PQLI** — Morris D. Morris (1979): literacy + IMR + life expectancy at age 1. **Income shamil nahi**, simple average.
- Variants: **IHDI (2010)**, **MPI**

2. Theories of Development

Theory	Engine of Growth	Core Mechanism	Ant / End
Classical	Capital accumulation	Diminishing returns → rent ↑ → profit ↓	Stationary State
Marx	Class conflict	Organic composition ↑ → profit rate ↓	Socialism (revolution)
Schumpeter	Innovation	Entrepreneur circular flow todta hai	Socialism (success se!)
Rostow	Investment ↑ + leading sectors	5 stages, take-off decisive	Mass consumption

> Classical (Smith, Ricardo, Malthus, Mill)

- **Smith: Division of Labour** — par "extent of the market" se limited
- **Ricardo:** land par diminishing returns → rent ↑, profit ↓ → stationary state
- **Malthus:** population **geometric**, food **arithmetic**
- **Mill:** stationary state **buri nahi** — mental/moral progress ka samay
- Assumptions: **Say's Law**, Wage Fund Theory, subsistence wage, laissez-faire
- **Sabse badi galti:** technological progress ko ignore kiya

> Marx - Historical Materialism

- Stages: Primitive Communism → Slavery → Feudalism → Capitalism → Socialism → Communism
- Economic **base** hi superstructure (law, politics, religion) ko determine karta hai

$$C = c + v + s \quad (c = \text{constant}, v = \text{variable/labour}, s = \text{surplus})$$

Rate of Surplus Value	Organic Composition	Rate of Profit
$s / v = \text{Rate of Exploitation}$	$c / (c + v)$	$s / (c + v)$

- Sirf **v (labour)** **naya value banata hai**; c sirf value transfer karta hai
- Chain: organic composition ↑ → **falling rate of profit** → **Reserve Army of Labour** → immiserization → concentration & centralisation → crises → revolution
- **Reproduction Schema** (Dept I = capital goods, Dept II = consumer goods) → **Feldman** → **Mahalanobis** → **India ki 2nd FYP**

> Schumpeter - Innovation Theory

- **Circular Flow** (profit = 0, interest = 0) ko **entrepreneur** todta hai
- **5 Innovations:** **naya product** • **nayi production method** • **naya market** • **naya raw-material source** • **naya organisation**
- **Invention ≠ Innovation** | **Entrepreneur ≠ Capitalist** — risk capitalist uthata hai
- Bank credit "**ex nihilo**" | Profit = **temporary**, imitators khatam kar dete hain
- Innovations "**swarm-like**" aate hain → boom → imitation → recession = **Creative Destruction**

TRAP: Schumpeter (1942) ke anusaar capitalism apni **SAFALTA (success)** se marega — innovation routinize ho gaya, entrepreneur obsolete. Ye **Marx** ke bilkul ulta reasoning hai.

> Rostow (1960) - "Stages of Economic Growth : A Non-Communist Manifesto"

- Traditional Society → Preconditions → **TAKE-OFF** → Drive to Maturity → High Mass Consumption

Take-off ki 3 conditions (guaranteed question):

(i) Investment **5% → 10%+ of NNP** (ii) **Leading sectors** with backward-forward linkages (iii) Supportive political-social-institutional framework

- Dating: Britain 1783–1802, USA 1843–60, Japan 1878–1900, **India 1952 se**
- **Kuznets** ne empirically challenge kiya — investment ratio double hone ka evidence nahi

3. Approaches to Development (sabse scoring!)

Approach	Economist & Year	Ek Line Mein
Vicious Circle of Poverty	Ragnar Nurkse (1953)	"A country is poor because it is poor" — supply + demand circle
Circular Cumulative Causation	Gunnar Myrdal (1957)	UDCs mein Backwash > Spread → inequality badhti hai
Big Push	Rosenstein-Rodan (1943)	Minimum quantum of investment — hawai jahaz ki ground speed
Balanced Growth	Nurkse / Rodan / Lewis	Sab industries saath — har ek doosre ka market ban jaye
Unbalanced Growth	A. O. Hirschman (1958)	Jaanbujhkar unbalance karo — linkages investment induce karenge
Critical Minimum Effort	Leibenstein (1957)	Effort critical size se bada ho, warna wapas trap mein
Low Level Equilibrium Trap	R. R. Nelson (1956)	Population growth income gains kha jaati hai (Malthusian)
Unlimited Supply of Labour	W. A. Lewis (1954)	Dual economy — surplus labour, Lewis Turning Point

> Zaroori Details

- Nurkse ke 2 circles: Supply side (low income → low saving → low investment) & Demand side (low income → **small market** → low inducement). Disguised unemployment = "**hidden saving potential**"
- Myrdal: **Backwash** (capital/labour rich region kheench le) vs **Spread** (fayda phaile). UDCs = "Soft State". Hirschman ke terms = Polarization & Trickle-down
- Big Push ki 3 Indivisibilities: (i) Production function / **SOC** (30–35% of investment) (ii) **Demand** (complementarity) (iii) **Supply of savings**
- Balanced Growth ki criticism: **H. W. Singer** — "agar resources hote to underdeveloped hi kyun hota?" | **Fleming** — demand side complementarity, par supply side competition
- Hirschman: UDCs mein kami **decision-making ability** ki hai, resources ki nahi. **Backward linkage** = input suppliers, **Forward linkage** = output users. Sabse zyada linkage = **intermediate goods (steel)**, sabse kam = **agriculture**
- Hirschman ke 2 paths: excess SOC = "**permissive**" | SOC shortage = "**compelling**" (UDCs ke liye behtar)
- Leibenstein vs Nelson: conclusion same (bada dhakka chahiye), par Nelson = graphical/mathematical trap, Leibenstein = broader socio-economic effort

> Lewis Dual Economy Model - detail

Subsistence / Traditional Sector	Capitalist / Modern Sector
MPL ≈ 0 (disguised unemployment)	MPL positive
Wage = Average Product (institutional)	Wage = MPL, subsistence se ~30% zyada

- Labour supply curve **perfectly elastic (horizontal)** → wage constant → poora fayda profit mein → reinvestment → surplus khatam → **LEWIS TURNING POINT** → wages badhne lagti hain
- Lewis ka statement: saving-investment ko **4–5% se 12–15% of NI** tak le jaana hi central problem hai. Extension = **Fei-Ranis**
- Criticism: **T. W. Schultz** — 1918–19 influenza epidemic data se MPL ≠ 0 sabit kiya | **Todaro** — urban unemployment pehle se hai
- Nobel: Myrdal (1974), Lewis & Schultz (1979), Leontief (1973), Kantorovich & Koopmans (1975)

4. Choice of Techniques & Cost-Benefit Analysis

Central conflict: Labour-intensive = **aaj ka employment / output** ↑ vs Capital-intensive = **kal ka surplus / growth** ↑

Investment Criterion	Economist	Kya Maximize	Technique
Capital Turnover	Polak & Buchanan	Output per unit capital (min. C-O ratio)	Labour-intensive
Social Marginal Productivity	Kahn & Chenery	Social output (<u>static</u>)	Neutral
MRQ / Reinvestment	Galenson & Leibenstein	<u>Reinvestable surplus</u>	Capital-intensive
Time Series	Otto Eckstein	Discounted growth stream	Compromise
Choice of Techniques	A. K. Sen	"Period of Recovery"	Short = LI, Long = CI

TRAP: Capital Turnover aur MRQ bilkul opposite hain. SMP reinvestment ignore karta hai (static), MRQ dynamic hai.

- Appropriate / Intermediate Technology — E. F. Schumacher, "Small is Beautiful" (1973), ITDG 1966
- Ratio per workplace: £1 (traditional) : £100 (intermediate) : £1,000 (Western)
- Motto: "Production by the masses, not mass production" (Gandhian). India: KVIC, Amul, bio-gas, MGNREGA
- CBA base: **Kaldor-Hicks Compensation Principle**

$$NPV = \sum (B_t - C_t) / (1 + r)^t > 0 \quad | \quad BCR > 1 \quad | \quad IRR > \text{social discount rate}$$

- Conflict ho to **NPV preferred** (multiple-IRR problem)
- Shadow prices: unskilled labour ka shadow wage ≈ 0 | foreign exchange ka shadow price **zyada** (rupee overvalued)

Aadhaar	Little-Mirrlees (OECD, 1969/74)	UNIDO (1972)
Authors	I. M. D. Little & J. A. Mirrlees	Dasgupta, Marglin & Sen
Numeraire	Uncommitted social income in <u>free foreign exchange</u>	<u>Aggregate consumption</u>
Prices	World / border prices	Domestic prices

5. Planning, NITI Aayog & SDGs

- Types: Imperative vs Indicative • Direction vs Inducement (Lewis) • Physical vs Financial • **Rolling Plan** = Myrdal ka suggestion, India 1978-80
- Techniques: Input-Output = **Leontief** • Linear Programming = **Kantorovich & Koopmans** • CBA • PERT/CPM • SAM = Richard Stone
- India ke models: 1st Plan = **Harrod-Domar** | 2nd Plan = **Mahalanobis** (heavy industry) | 8th Plan (1992-97) = Indicative planning ki shuruat
- Indicative planning ka janmdata = FRANCE ("Planification", Jean Monnet, 1946). Market economy mein State = facilitator, regulator, enabler
- **12th Plan (2012-17) = aakhri Five Year Plan**. Ab: 15-yr Vision, 7-yr Strategy, 3-yr Action Agenda

Aadhaar	Planning Commission (1950)	NITI Aayog (1 Jan 2015)
Approach	Top-down, directive	<u>Bottom-up, advisory think tank</u>
Fund allocation	Haan — States ko funds allot karta tha	NAHI — ab Finance Ministry ke paas
Status	Cabinet Resolution	Na Constitutional, na Statutory — sirf Cabinet Resolution
Body	NDC (National Devp. Council)	Governing Council = sabhi CM + UT ke LG

- Structure: Chairperson = **PM** • Vice Chairman = **Ashok Lahiri** (2026-) • CEO = **Nidhi Chhibber** (2026-)
- VC ka kram: Panagariya (2015-17) → Rajiv Kumar (2017-22) → Suman Bery (2022-26) → Ashok Lahiri
- Initiatives: SDG India Index • Aspirational Districts (112, Jan 2018) & Blocks (500, 2023) • Atal Innovation Mission •

> Sustainable Development & SDGs

Brundtland Commission — "Our Common Future" (1987): aisa development jo vartaman peedhi ki zarooratein पूरी kare, bhavishya ki peedhi ki kshamta se samjhauta kiye bina.

Year	Milestone
1972	Stockholm Conference + Club of Rome ki "Limits to Growth"
1987	Brundtland Report — "Our Common Future"
1992	Rio Earth Summit — Agenda 21, UNFCCC, CBD
2000	MDGs — 8 goals, 21 targets (2000–2015)
2015	SDGs adopted — Sept 2015, UNGA Resolution 70/1

- Document: "Transforming Our World : The 2030 Agenda for Sustainable Development"
- **17 Goals, 169 Targets, ~231 indicators** • 2015–2030 • 193 countries
- Motto: "Leave No One Behind" | **5 Ps = People, Planet, Prosperity, Peace, Partnership**

Aadhaar	MDGs (2000-15)	SDGs (2015-30)
Goals	8 goals, 21 targets	17 goals, 169 targets
Coverage	Sirf developing countries	Sabhi desh (universal)
Focus	Social sector	Social + Economic + Environmental
Process	Top-down (experts)	Consultative (193 desh)

- India ka framework: **NITI Aayog** = **nodal agency** • MoSPI = **NIF, 277 indicators** • Panchayati Raj = Localisation (LIF, Panchayat Advancement Index) • **VNR** to UN HLPF
- **SDG India Index:** pehla edition **Dec 2018**; India pehla desh jisne sub-national SDG monitoring institutionalise kiya. Score: **57 (2018) → 66 (2020-21) → 71 (2023-24)**
- **Global SDG Index (UN SDSN, 2026):** India rank **94 / 167**, score 68.3

Confusion mat karna: SDG India Index = NITI Aayog, states ki ranking, score 71 ≠ **Global SDG Index** = UN SDSN, desh ki ranking, India 94/167.

12 Confusion Traps – Ek Baar Aur

GALAT	SAHI
HDI arithmetic mean	Geometric mean (2010 se)
GDI ek inequality index hai	Ratio hai — Female HDI / Male HDI
GII mein zyada value = achha	Kam value = achha (0 = equality)
GII mein income شامل hai	Nahi — LFPR hai
Big Push = Balanced Growth	Big Push = size (kitna), Balanced = pattern (kahan)
Nurkse = unbalanced growth	Nurkse = Balanced, Hirschman = Unbalanced
Capital Turnover → capital-intensive	Capital Turnover → labour-intensive
MRQ → labour-intensive	MRQ (Galenson-Leibenstein) → capital-intensive
Little-Mirrlees = domestic prices	L-M = world prices; UNIDO = domestic
Schumpeter: risk entrepreneur uthata hai	Capitalist uthata hai
NITI Aayog statutory / constitutional body	Sirf Cabinet Resolution
NITI Aayog funds allot karta hai	Nahi — Finance Ministry karta hai

— Day 03 Complete. Next : Unit 4 – Growth Economics —